
Experiment no 11:
D3.js Lab: Building an Animated Bar Chart from Scratch

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1. **Requirements**
   VS Code
   Chrome Browser
   Live Server Extension
2. **Create Project Structure**
   Create a folder and files as follows:
   d3-lab
   index.html
   style.css
   script.js
5. **Step 1 Basic HTML Setup**
   Create index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>D3 Bar Chart</title>
   <script src="https://d3js.org/d3.v7.min.js"></script>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <h1>Bar Chart Race</h1>
   <div id="chart"></div>
   <script src="script.js"></script>
</body>
</html>
```

   Run using Live Server. The page should display only the heading.
6. **Step 2 Basic Styling**
   Create style.css

```
body {
   font-family: Arial;
   text-align: center;
}
#chart {
   margin-top: 30px;
}
```

7. **Step 3 Add Dataset**
   Open script.js and add

```
const data = [
   {year: 2020, name: "A", value: 50},
```

```
    {year: 2020, name: "B", value: 30},
    {year: 2020, name: "C", value: 20},
    {year: 2021, name: "A", value: 60},
    {year: 2021, name: "B", value: 40},
    {year: 2021, name: "C", value: 25},
    {year: 2022, name: "A", value: 80},
    {year: 2022, name: "B", value: 50},
    {year: 2022, name: "C", value: 35}
];
```

8.Step 4 Create SVG

```
const width = 600;
const height = 400;
const svg = d3.select("#chart")
    .append("svg")
    .attr("width", width)
    .attr("height", height);
```

9. Step 5 Create Scales

```
const xScale = d3.scaleLinear()
    .range([0, width - 100]);

const yScale = d3.scaleBand()
    .range([0, height])
    .padding(0.2);
```

10. Step 6 Create Update Function

```
function update(year) {
    const yearData = data.filter(d => d.year === year);
    xScale.domain([0, d3.max(yearData, d => d.value)]);
    yScale.domain(yearData.map(d => d.name));
    const bars = svg.selectAll("rect")
  .data(yearData, d => d.name);
  bars.enter()
  .append("rect")
       .attr("y", d => yScale(d.name))
       .attr("height", yScale.bandwidth())
       .attr("x", 0)
       .attr("width", 0)
       .attr("fill", "steelblue")
       .merge(bars)
       .transition()
```

```
        .duration(1000)
        .attr("y", d => yScale(d.name))
        .attr("width", d => xScale(d.value));

    bars.exit().remove();
}
```

11. **Step 7 Call Function**

```
update(2020);
```

Bars should now be visible.

12. **Step 8 Add Animation**

```
const years = [2020, 2021, 2022];
let i = 0;
setInterval(() => {
    update(years[i]);
    i = (i + 1) % years.length;
}, 1500);
```

Bars will animate automatically.

13. **Step 9 Add Labels**

Add this inside the update function after bars code

```
const labels = svg.selectAll("text")
    .data(yearData, d => d.name);

labels.enter()
    .append("text")
    .attr("y", d => yScale(d.name) + 20)
    .attr("x", 5)
    .text(d => d.name)
    .merge(labels)
    .transition()
    .attr("y", d => yScale(d.name) + 20);

labels.exit().remove();
```

14. **Output**

An animated bar chart should be visible with changing values across years.

15. **Practice Tasks**

Change dataset values
Change bar color by modifying fill attribute
Change animation speed by reducing duration
Add one more category to dataset